Unnamed_Unit

Unite 1 - Conditional Statements

PDF Documents

PDF Document 1: 1734182943784-Chapter 3_unit1 Conditional Statements.pdf

This PDF document is included in the unit materials.

UNIT 1 CONDITIONAL STATEMENTS

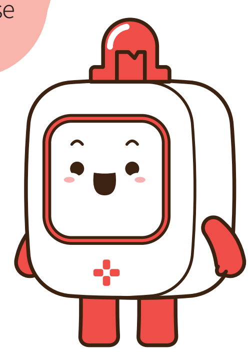
If Statement

Imagine you're deciding whether to carry an umbrella outside. The simple rule you might follow is: "If it's raining, then I'll take an umbrella. If it's not raining, I won't need one."

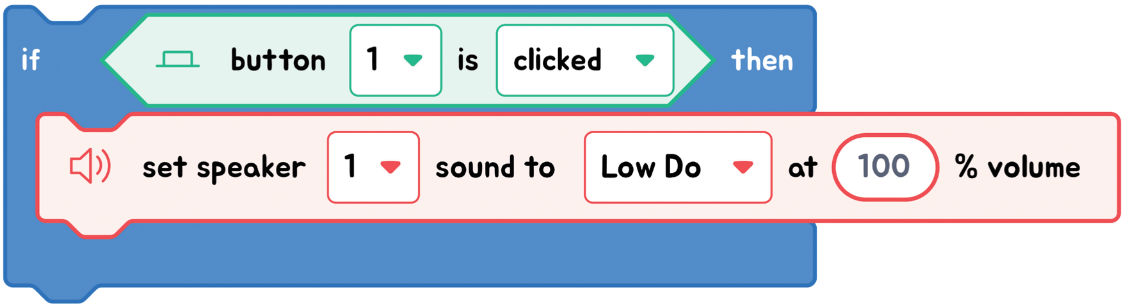
This decision-making process is similar to how an "if statement" functions in programming. It acts like a rule or condition, essentially saying, "If a certain condition is true, then perform a specific action."

THINK ABOUT IT!

Think of another everyday situation where an "if" rule could apply. How would you phrase it?



EXPLANATION WITH CODE EXAMPLE



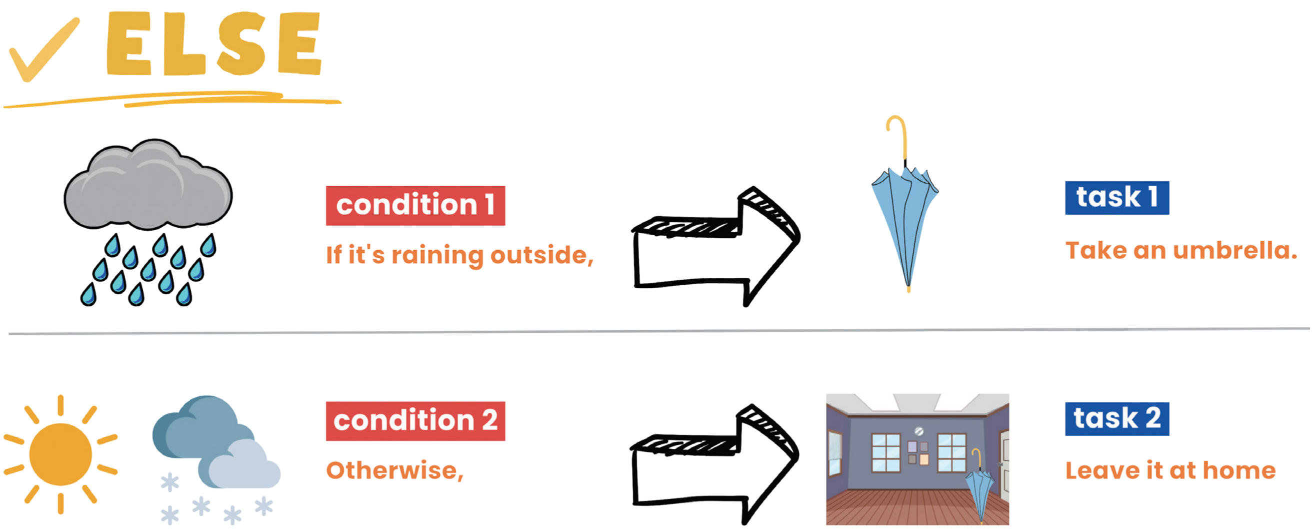
In block coding, we use the "if statement" to check conditions. If the condition is met (or true), the program will carry out a specific task. The "if statement" block visually separates the condition from the action with "if" and "then" parts. It examines the condition, and if the condition holds true, it proceeds to execute the task inside the block.

Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the tasks inside the "if" statement are carried out. These tasks specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."

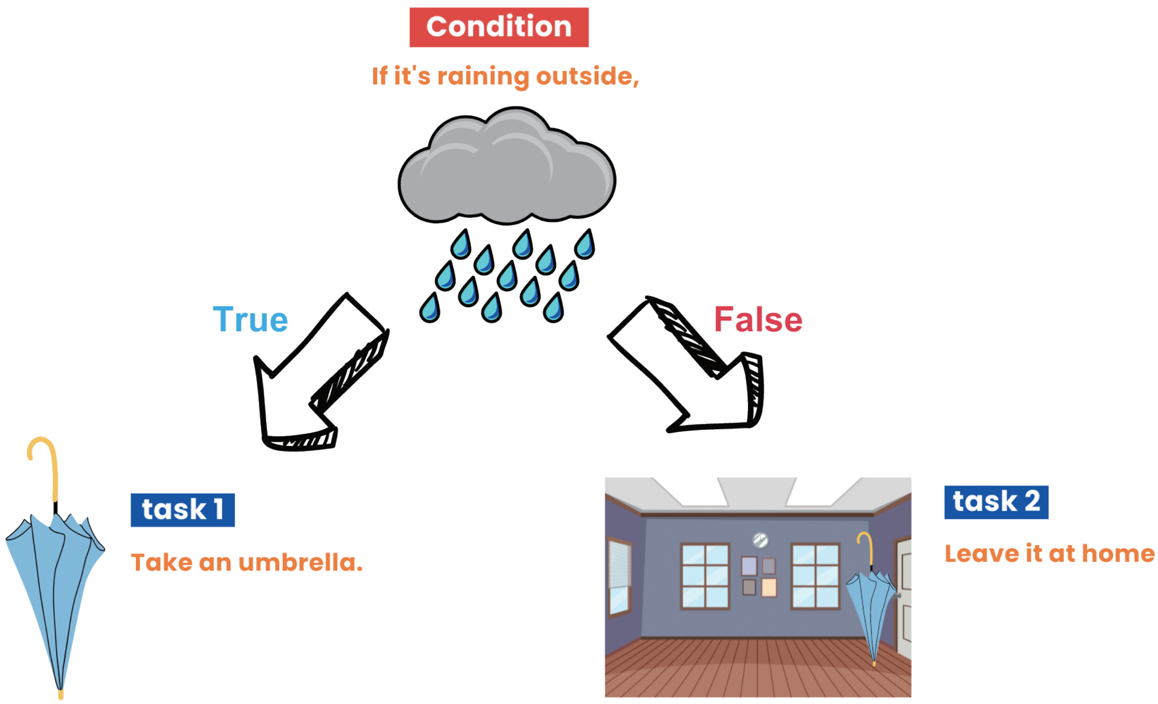
If else Statement

An 'If-Else' statement is like an 'If' statement with a backup plan. If the 'If' part isn't true, then the 'Else' part takes over and does something else instead.



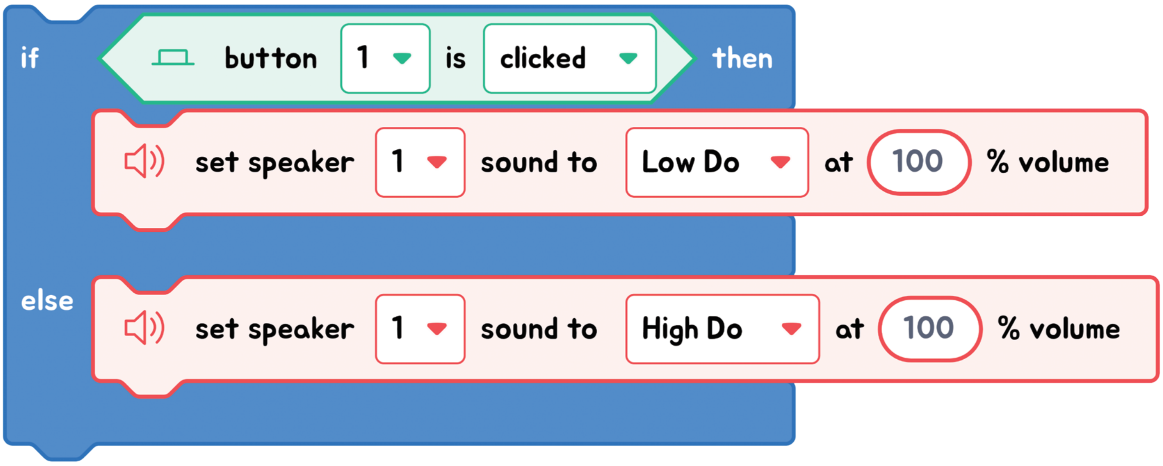
Imagine you're deciding whether to bring an umbrella. An 'If-Else' statement can help you with this!

If it's raining outside (this is the condition), you'll take an umbrella (this is the task). Otherwise (this is the Else part), you'll leave the umbrella at home. So, even though we only consider rain, Else takes care of everything else by default.

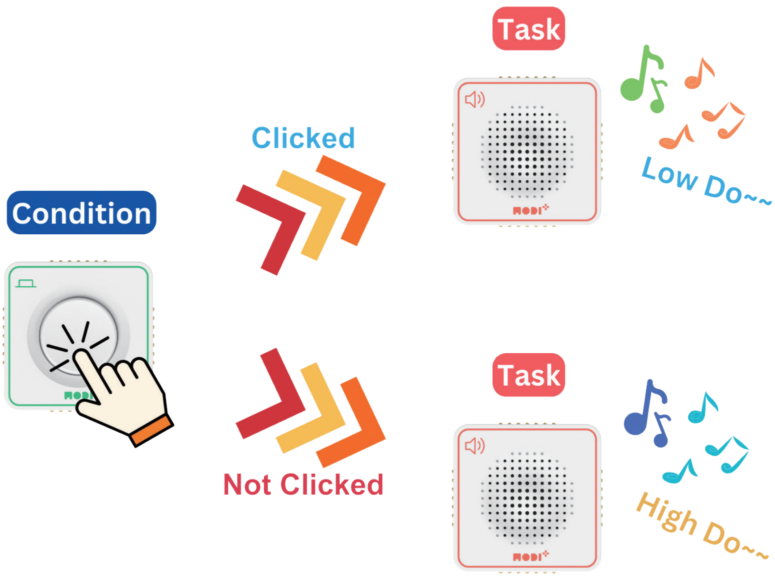


The key to 'If-Else' is that if the 'If' condition isn't met, the program will do what's in the 'Else' part instead.

EXPLANATION WITH CODE EXAMPLE



In block coding, the "If-Else" statement lets us choose between two actions based on a condition. If the condition is true, one action is taken; if not, a different action is executed. The "If-Else" block visually splits this choice, ensuring the program always has a clear direction, no matter the condition's outcome.

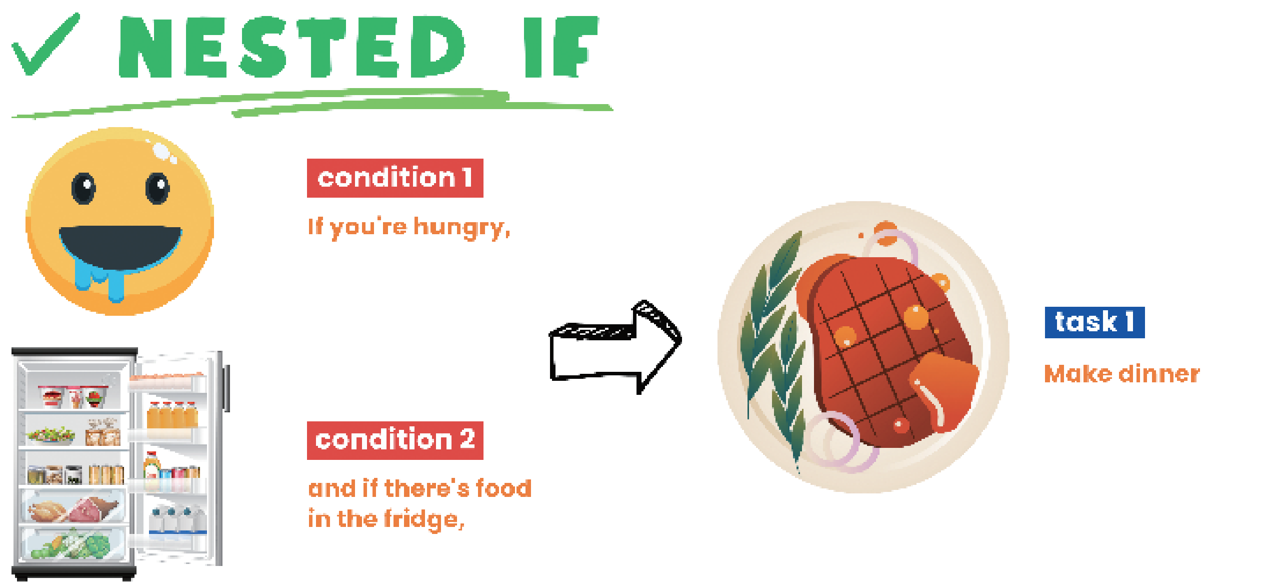


Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the instructions inside the "if" statement are carried out. These instructions specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."
3. **Execution if False:** If the condition is not met, then the instructions inside the "else" statement are carried out. These instructions specify the actions to take when the condition is not met. Thus, the task performed is "Set the speaker to play the 'high do' sound at 100% volume."

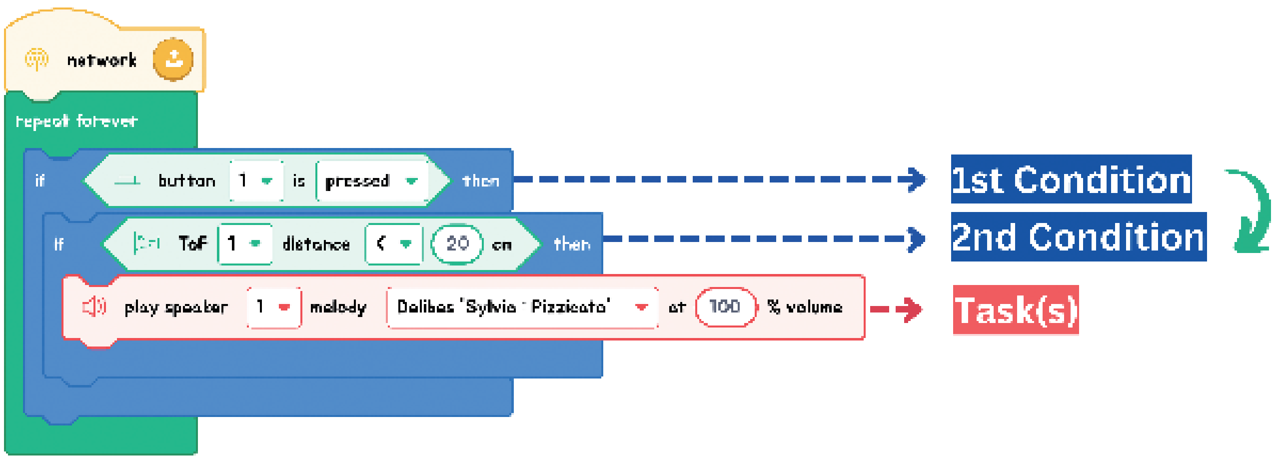
Nested If Statement

Nested if statements are like putting one decision inside another. First, you check a condition; if that's true, you check another one inside it. If this second condition is also true, then an action takes place. It's like solving a layered puzzle: solve one layer to get to the next!



First, you check if you're hungry (1st condition). If you are, you then look inside the fridge to see if there's food (2nd condition). If there's food available, you go ahead and make dinner (this is the task).

So, with a 'Nested If', you first decide if you're hungry, and only then do you consider whether there's food in the fridge to make a meal. Let's look at the example below!

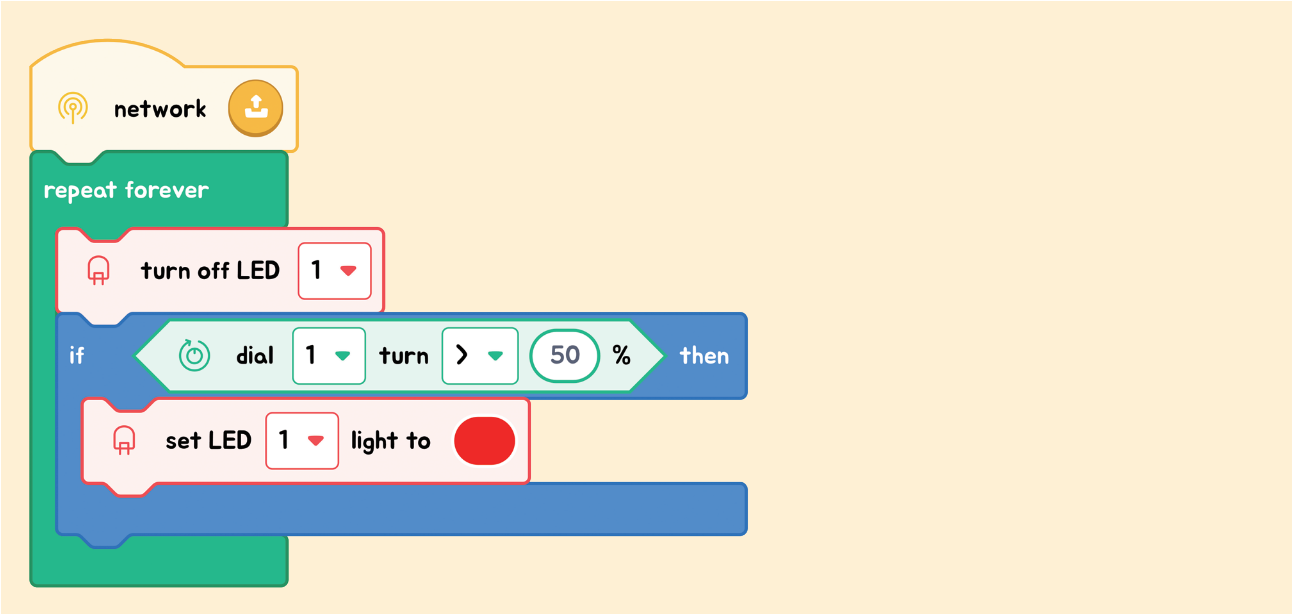


First, the program checks if a button is pressed (1st condition). If the button is indeed pressed, it then examines another condition: whether the distance measured by the ToF sensor is less than 20 cm (2nd condition). Only if both these conditions are met, does the program command the speaker to play the song.

Exercise

Exercise 1

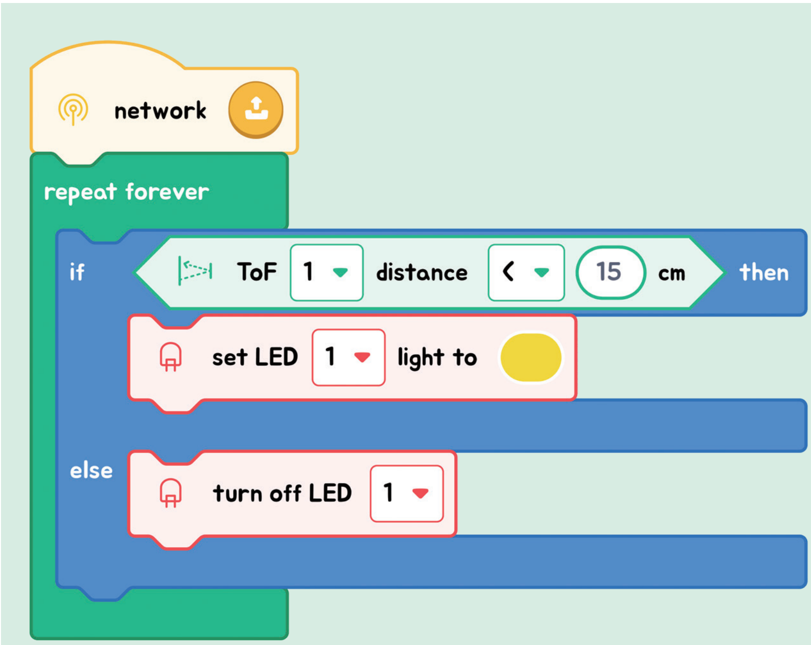
Change this code to an **'If-Else' statement** in the Code Editor.



Exercise 2

Use the Code Editor to create the following project:

"We're making a sensor light. If an object is close to the sensor, the light will turn on. If there's no object nearby, the LED will be off."



TIPS

By monitoring with the ToF (Time of Flight) sensor, determine the distance at which you want the light to turn on.

Exercise 3

3.1 Based on the following scenario, fill in the table with the appropriate information.

Scenario

Using the Environment module, we're controlling a fan based on temperature and humidity readings. If the temperature exceeds 25 degrees Celsius and the humidity is more than 60%, the fan will turn on at maximum speed. If these conditions are not met, the fan will remain off.

Condition 1	
Condition 2	
Task	

3.2 Using block coding, match the necessary blocks to implement the logic described in the table above.

Ex) If the temperature exceeds 25 degrees Celsius,	Check the other condition

TIPS

We use 'nested if statements' when we need to check several things before making a decision.
The actions in these statements happen only if all the conditions are true